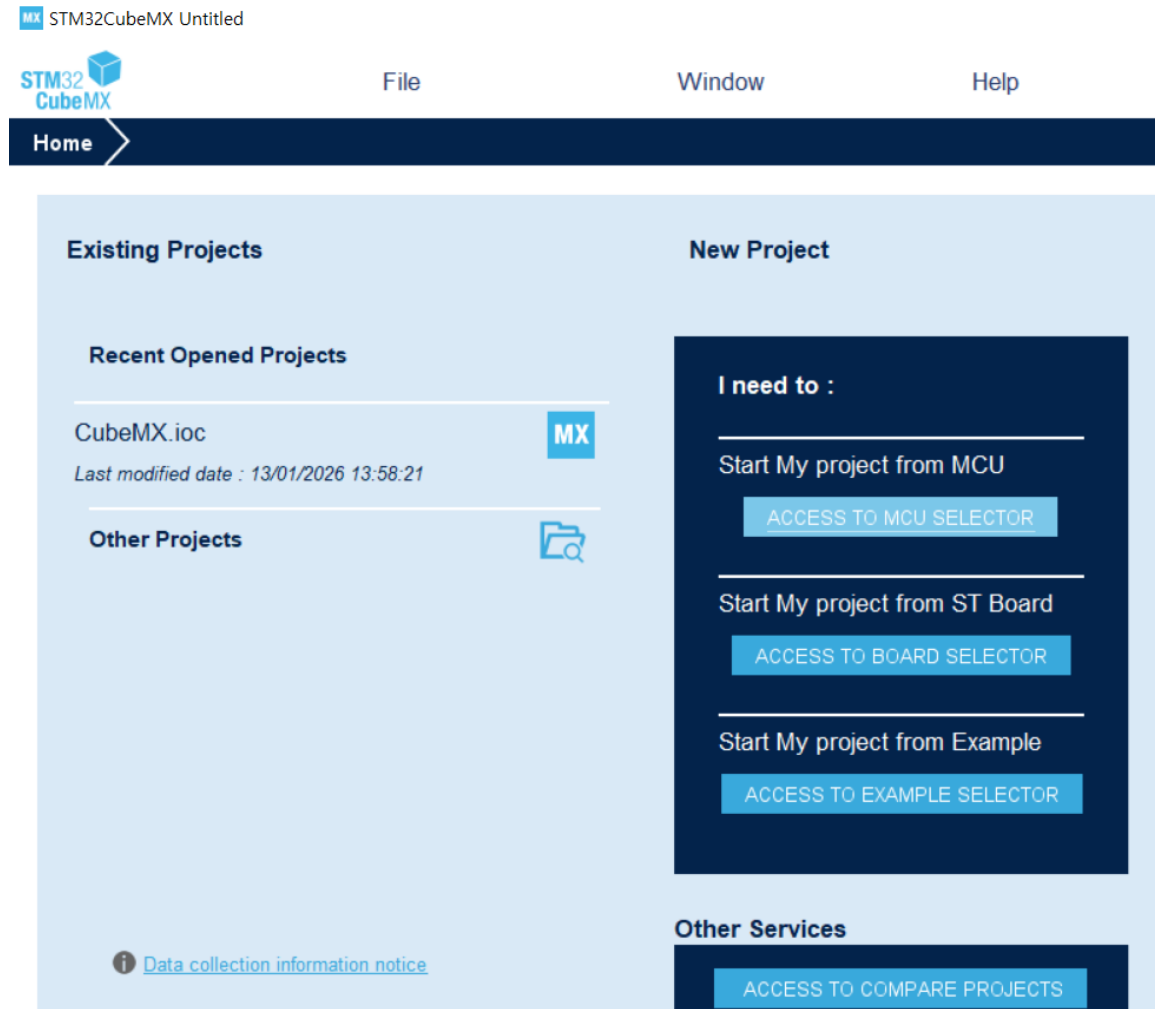


BN0055 – 9축 센서 실습

2026.06.17

Project 생성

- STM32CubeMX 실행 후 "ACCESS TO MCU SELECTOR" 선택



Project 생성

- STM32F407ZET6 선택 후 "Start Project" 클릭

MX New Project from a MCU/MPU

MCU/MPU Selector Board Selector Example Selector Cross Selector

MCU/MPU Filters

Commercial Part Number STM32F407ZET6

PRODUCT INFO

- Segment
- Series
- Line
- Marketing Status
- Price
- Package
- Core
- Coprocessor

STM32F4 Series

STM32F407ZET6

High-performance foundation line, Arm Cortex-M4 core with DSP and FPU, 512 Kbytes of Flash memory, 168 MHz CPU, ART Accelerator, Ethernet, FSMC

ACTIVE
Product is in mass production

Unit Price for 10kU (US\$): 5.605

4 mm LQFP 144 20x20x1.

The STM32F405xx and STM32F407xx family is based on the high-performance

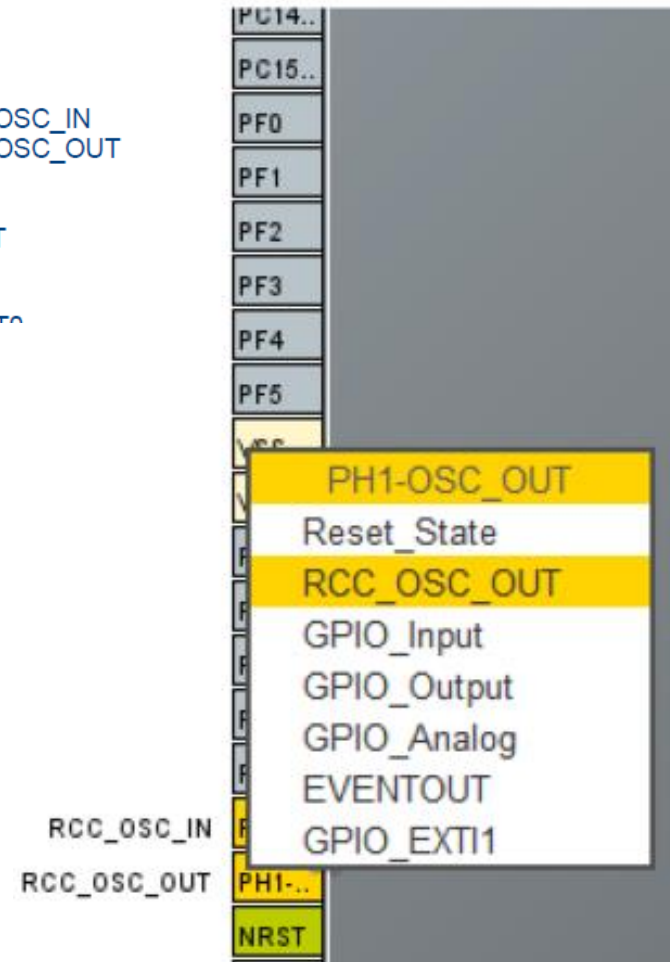
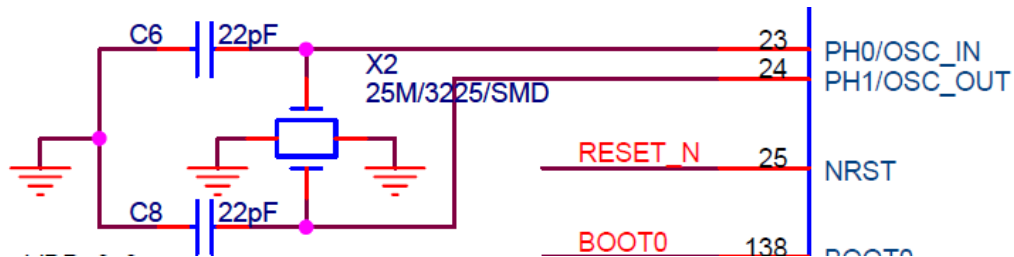
MCUs/MPUs List: 2 items

* Comm...	Part No	Reference	Mark...	Unit Pric...	B...	Package	Flash	RAM	...	Fre...
☆ STM32F4...	STM32...	STM32...	Active	5.605		LQFP 144 20x...	512 ...	192 ...	1...	168 ...
☆ STM32F...	STM32...	STM32...	Active	5.605		LQFP 144 20x...	512 ...	192 ...	1...	168 ...

Export

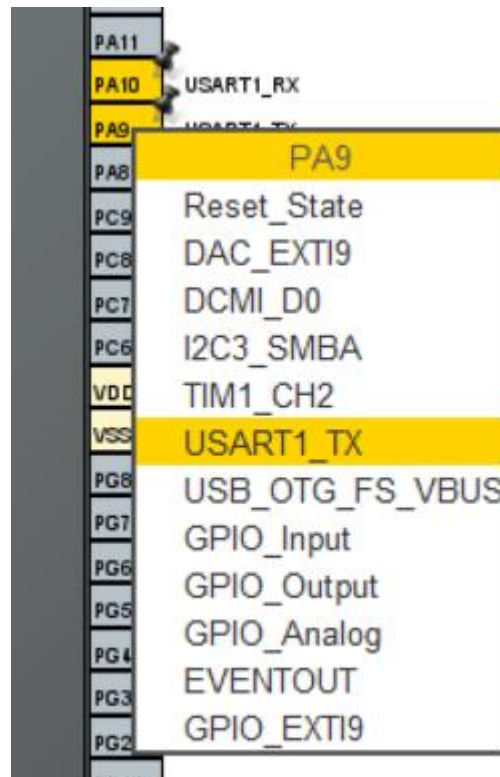
Project 생성

- HSE(25MHz) 핀 설정 - 회로도 참조
- HSE : PH0-OSC_IN, PH1-OSC_OUT



Project 생성

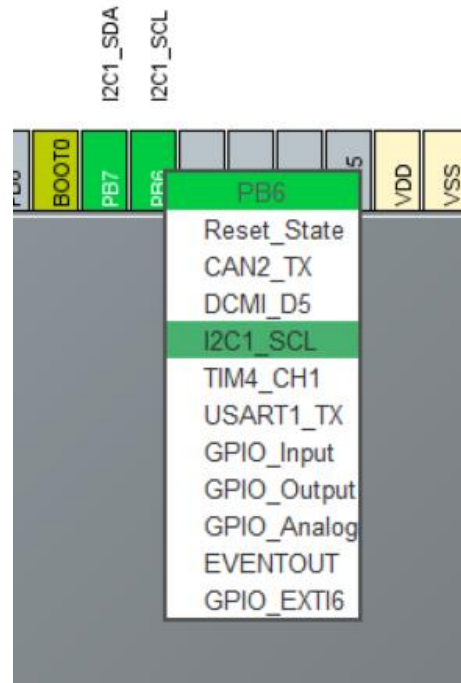
- USART1핀 설정 - 회로도 참조
- UART1 : PA9-USART1_TXD, PA10-USART1_RXD



Project 생성

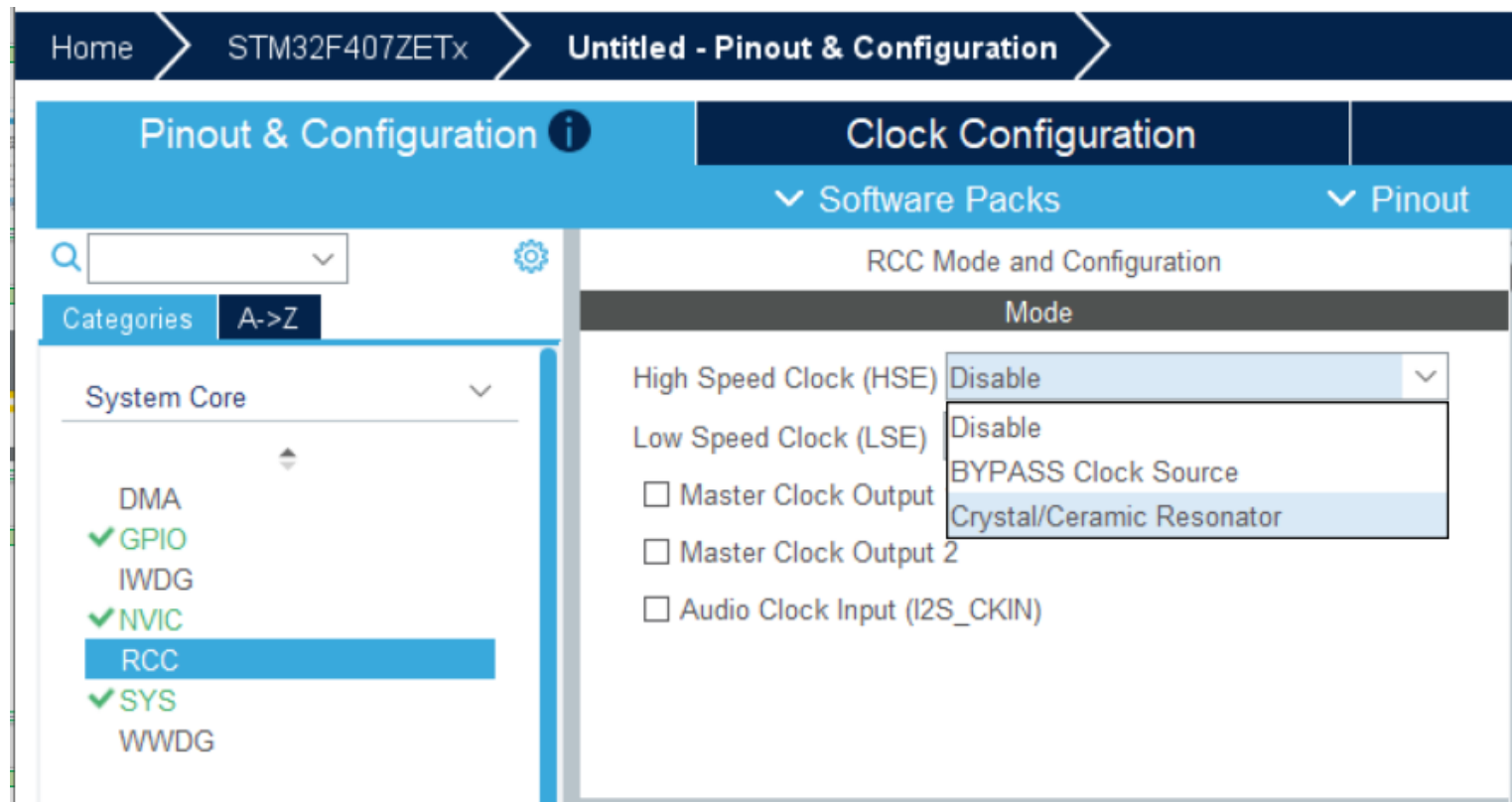
- I2C 핀 설정 - 회로도 참조
- I2C : PB6-I2C1_SCL, PB7-I2C1_SDA

PB5	136	I2C1_SCL
PB6	137	I2C1_SDA
PB7	139	CAN1_RX



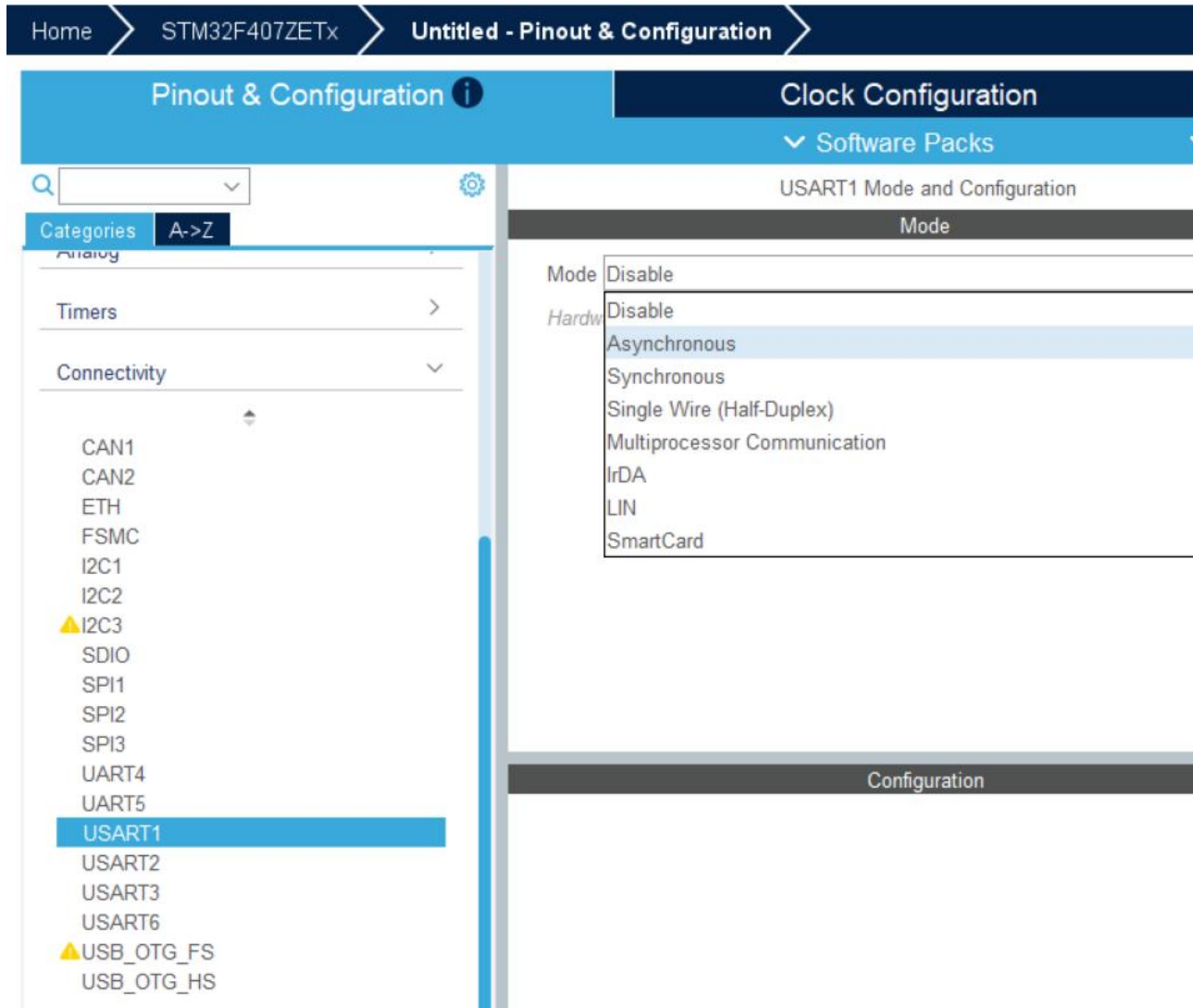
Project 생성

- System Core -> RCC에서 HSE 활성화 -> Crystal/Ceramic Resonator 선택



Project 생성

- Connectivity -> USART1 -> Asynchronous 선택



Project 생성

- Connectivity -> I2C1 -> Mode -> I2C 선택

The screenshot shows the STM32CubeMX interface for configuring the I2C1 peripheral. On the left, the 'Connectivity' category is expanded, and 'I2C1' is selected. The main window is titled 'I2C1 Mode and Configuration' and is divided into two main sections: 'Mode' and 'Configuration'.

Mode Section:

- A dropdown menu shows 'I2C' selected.

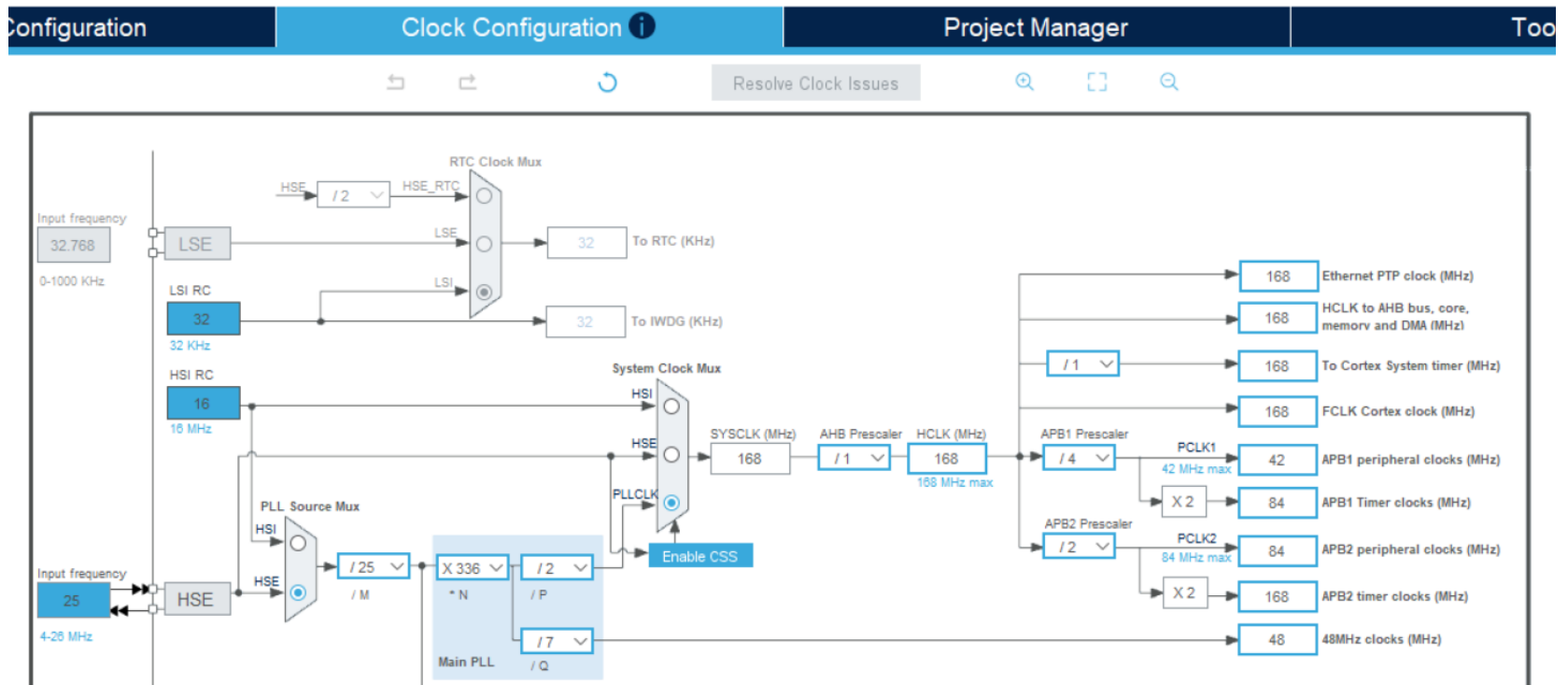
Configuration Section:

- A 'Reset Configuration' button is present.
- Four tabs are visible: 'NVIC Settings', 'DMA Settings', 'GPIO Settings', and 'Parameter Settings' (which is currently selected).
- Below the tabs, a text box says 'Configure the below parameters :'.
- A search bar with the placeholder 'Search (Ctrl+F)' is located above the parameter list.
- The parameters are organized into two sections: 'Master Features' and 'Slave Features'.

Feature	Value
Master Features	
I2C Speed Mode	Standard Mode
I2C Clock Speed (Hz)	100000
Slave Features	
Clock No Stretch Mode	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0
General Call address detection	Disabled

Project 생성

- Clock Configuration 클릭 -> Yes 클릭



Project 생성

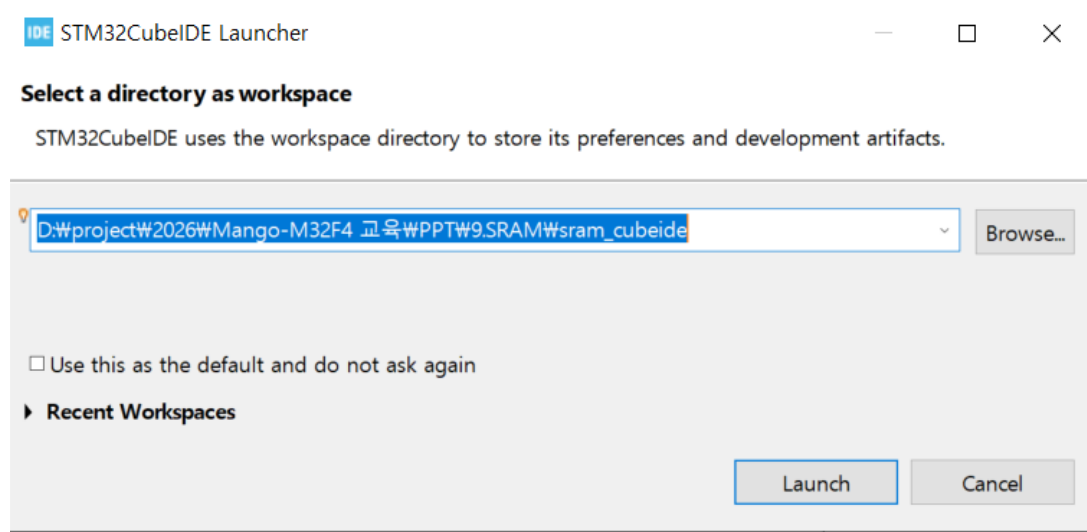
- Project Manager에서 Project Name 기입, IDE를 STM32CubeIDE로 선택 후 GENERATE CODE 클릭

The screenshot shows the STM32CubeIDE Project Manager interface. The top navigation bar includes 'Home', 'STM32F407ZETx', 'Untitled - Project Manager', and a 'GENERATE CODE' button. Below this, there are four tabs: 'Pinout & Configuration', 'Clock Configuration', 'Project Manager' (which is active), and 'Tools'. On the left side of the 'Project Manager' tab, there are two sub-tabs: 'Project' and 'Code Generator'. The 'Project' sub-tab is selected, displaying 'Project Settings'. The settings include: 'Project Name' (sram_cubemx), 'Project Location' (D:\project\2026\Mango-M32F4 교육\PPT\9.SRAM\), 'Application Structure' (Advanced), 'Toolchain Folder Location' (D:\project\2026\Mango-M32F4 교육\PPT\9.SRAM\), and 'Toolchain / IDE' (STM32CubeIDE). A checkbox for 'Generate Under Root' is checked.

Project Settings	
Project Name	sram_cubemx
Project Location	D:\project\2026\Mango-M32F4 교육\PPT\9.SRAM\ Browse
Application Structure	Advanced Do not generate the main()
Toolchain Folder Location	D:\project\2026\Mango-M32F4 교육\PPT\9.SRAM\
Toolchain / IDE	STM32CubeIDE Generate Under Root

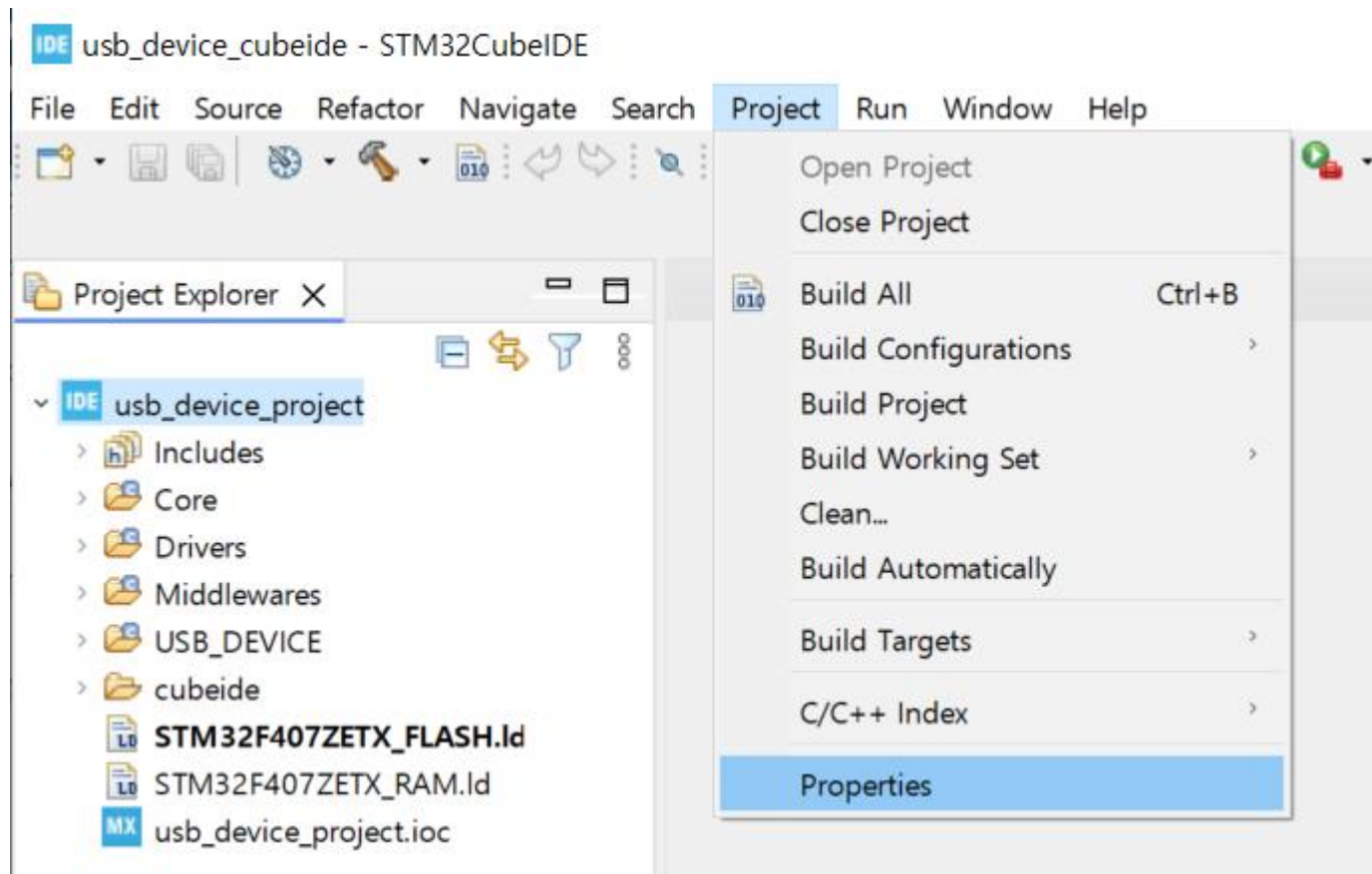
Project 생성

- Open Project 클릭하여 STM32CubeIDE 실행, workspace는 CubeMX와 다른 폴더로 지정



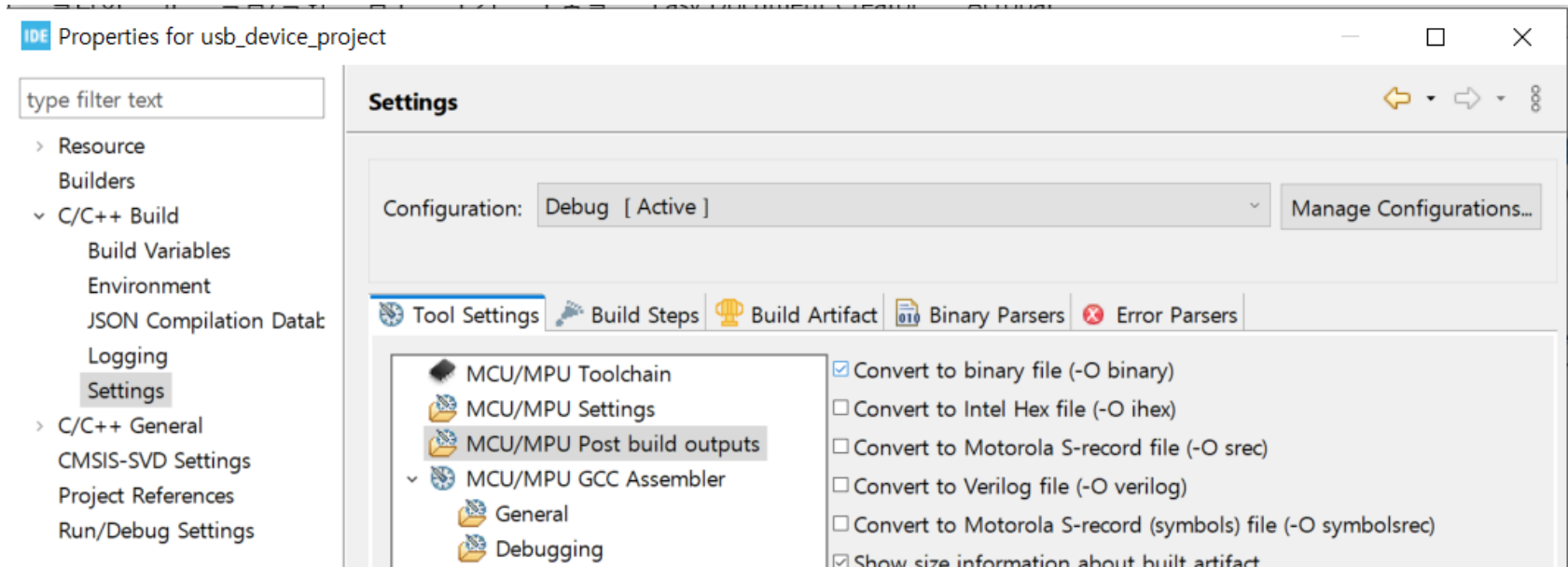
BIN 파일 생성 설정

- STM32CubeProgrammer로 시리얼 다운로드하기 위해서는 BIN파일이 필요.
- Project -> Properties 클릭



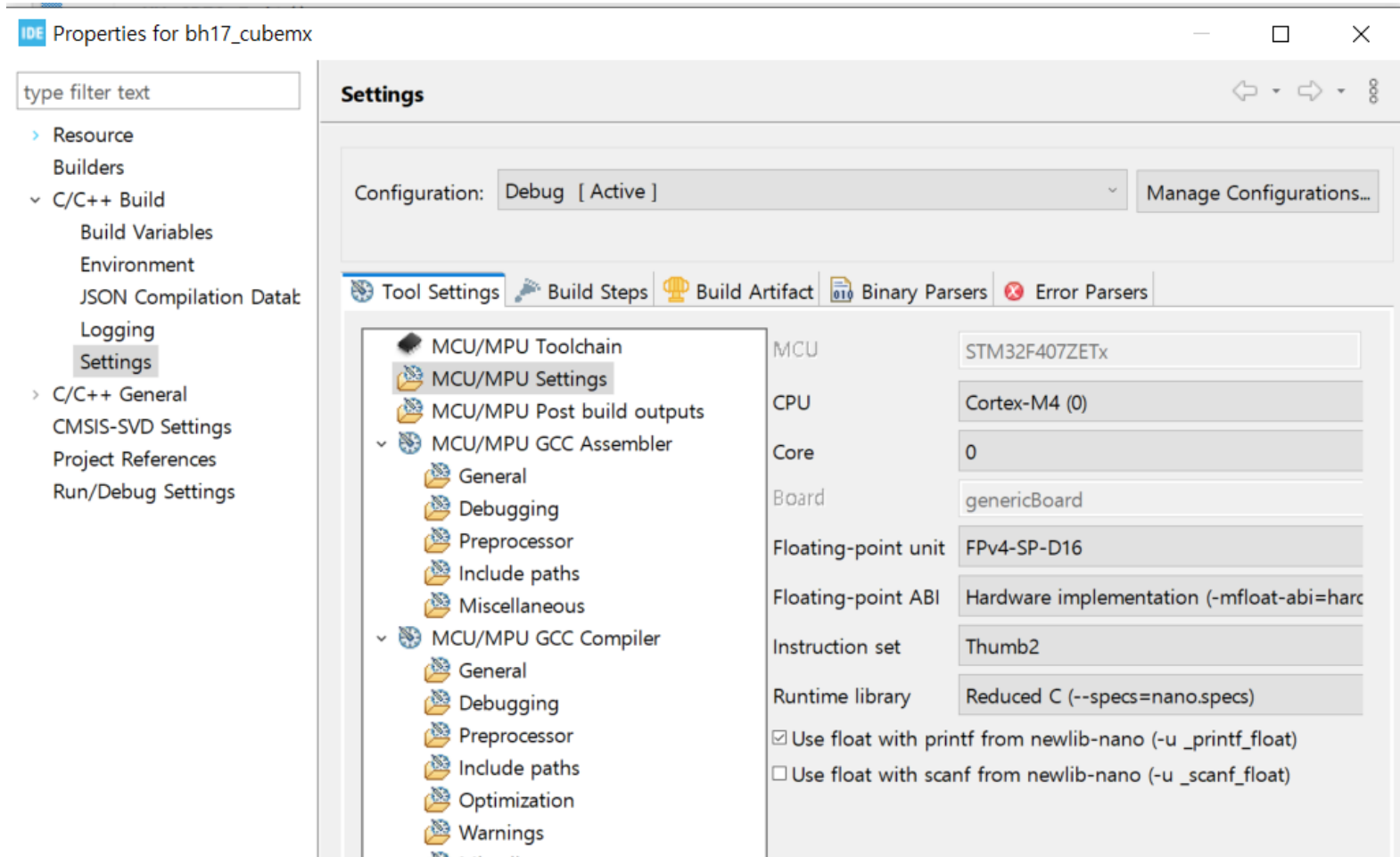
BIN 파일 생성 설정

- C/C++ Build -> Settings -> Tool Settings -> MCU/MPU Post build outputs => Convert to binary file(-O binary) 클릭.



빌드 설정

- C/C++ Build -> Settings -> Tool Settings -> MCU/MPU Settings => Use float with printf from newlib-nano(-u _printf_float) 클릭.



BNO055 데이터시트

- Slave Address : 0x28

Table 4-7: I2C address selection

I2C configuration	COM3_state	I2C address
Slave	HIGH	0x29
Slave	LOW	0x28
HID-I2C	X	0x40

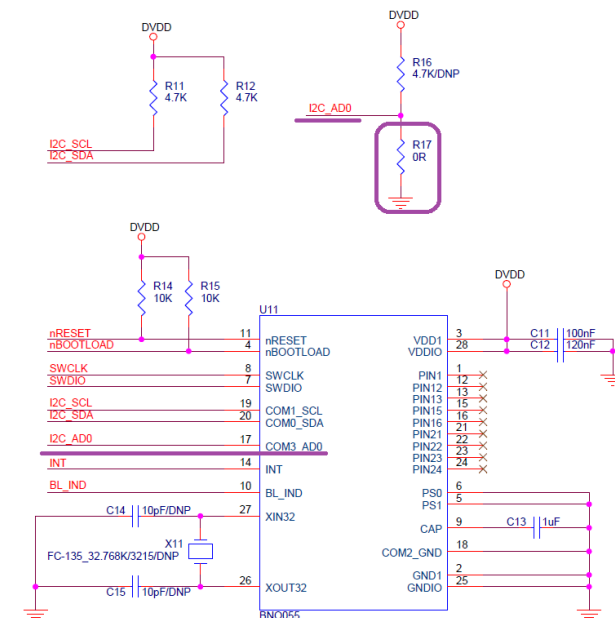
- CHIP_ID_REG : 0x00

4.3.1 CHIP_ID 0x00

	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Access	r	r	r	r	r	r	r	r
Reset	1	0	1	0	0	0	0	0
Content	BNO055 CHIP ID							

DATA	bits	Description
BNO055 CHIP ID	<7:0>	Chip identification code, read-only fixed value 0xA0

9 AXIS_SENSOR



- S/W reset : 0x3F, 0x20

4.3.63 SYS_TRIGGER 0x3F

	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Access	r/w	r/w	w	r/w	r/w	r/w	r/w	w
Reset	0	0	0	0	0	0	0	0
Content	CLK_SEL	RST_INT	RST_SYS	Reserved				Self_Test

DATA	bits	Description
CLK_SEL	7	0: Use internal oscillator 1: Use external oscillator. Set this bit only if external crystal is connected
RST_INT	6	Set to reset all interrupt status bits, and INT output
RST_SYS	5	Set to reset system
Self_Test	0	Set to trigger self-test

- OPR Mode : 0x3D

4.3.61 OPR_MODE 0x3D

	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Access	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w
Reset	0	0	0	1	0	0	0	0
Content	Reserved				Operation Mode <3:0>			

DATA	bits	Description
Operation Mode <3:0>	<3:0>	Read: Current selected operation mode Write: Select operation mode See Table 3-5 for details

Table 3-5: operating modes selection

Parameter	Value	[Reg Addr]: Reg Value
CONFIG MODE	CONFIGMODE	[OPR_MODE]: xxxx0000b
Non-Fusion Mode	ACCONLY	[OPR_MODE]: xxxx0001b
	MAGONLY	[OPR_MODE]: xxxx0010b
	GYROONLY	[OPR_MODE]: xxxx0011b
	ACCMAG	[OPR_MODE]: xxxx0100b
	ACCGYRO	[OPR_MODE]: xxxx0101b
	MAGGYRO	[OPR_MODE]: xxxx0110b
	AMG	[OPR_MODE]: xxxx0111b
Fusion Mode	IMU	[OPR_MODE]: xxxx1000b
	COMPASS	[OPR_MODE]: xxxx1001b
	M4G	[OPR_MODE]: xxxx1010b
	NDOF_FMC_OFF	[OPR_MODE]: xxxx1011b
	NDOF	[OPR_MODE]: xxxx1100b

- Power Mode : 0x3E

4.3.62 PWR_MODE 0x3E

	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Access	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w
Reset	0	0	0	0	0	0	0	0
Content	Reserved					Power Mode <1:0>		

DATA	bits	Description
Power Mode <1:0>	<1:0>	Read: Current selected power mode Write: Select power mode See Table 3-1 for details

Table 3-1: power modes selection

Parameter	Value	[Reg Addr]: Reg Value
Power Mode	Normal Mode	[PWR_MODE]: xxxxxx00b
	Low Power Mode	[PWR_MODE]: xxxxxx01b
	Suspend Mode	[PWR_MODE]: xxxxxx10b
	Invalid	[PWR_MODE]: xxxxxx11b

- NDOF Mode : 0x3D

4.3.61 OPR_MODE 0x3D

	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Access	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w
Reset	0	0	0	1	0	0	0	0
Content	Reserved				Operation Mode <3:0>			

DATA	bits	Description
Operation Mode <3:0>	<3:0>	Read: Current selected operation mode Write: Select operation mode See Table 3-5 for details

Table 3-3: Operating modes overview

Operating Mode		Available sensor signals			Fusion Data	
		Accel	Mag	Gyro	Relative orientation	Absolute orientation
Non-fusion modes	CONFIGMODE	-	-	-	-	-
	ACCONLY	X	-	-	-	-
	MAGONLY	-	X	-	-	-
	GYROONLY	-	-	X	-	-
	ACCMAG	X	X	-	-	-
	ACCGYRO	X	-	X	-	-
	MAGGYRO	-	X	X	-	-
Fusion modes	AMG	X	X	X	-	-
	IMU	X	-	X	X	-
	COMPASS	X	X	-	-	X
	M4G	X	X	-	X	-
	NDOF_FMC_OFF	X	X	X	-	X
	NDOF	X	X	X	-	X

• Power Mode : 0x3E

4.3.62 PWR_MODE 0x3E

	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Access	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w
Reset	0	0	0	0	0	0	0	0
Content	Reserved					Power Mode <1:0>		

DATA	bits	Description
Power Mode <1:0>	<1:0>	Read: Current selected power mode Write: Select power mode See Table 3-1 for details

Table 3-1: power modes selection

Parameter	Value	[Reg Addr]: Reg Value
Power Mode	Normal Mode	[PWR_MODE]: xxxxxx00b
	Low Power Mode	[PWR_MODE]: xxxxxx01b
	Suspend Mode	[PWR_MODE]: xxxxxx10b
	Invalid	[PWR_MODE]: xxxxxx11b

• NDOF Mode : 0x3D

4.3.61 OPR_MODE 0x3D

	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Access	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w
Reset	0	0	0	1	0	0	0	0
Content	Reserved				Operation Mode <3:0>			

DATA	bits	Description
Operation Mode <3:0>	<3:0>	Read: Current selected operation mode Write: Select operation mode See Table 3-5 for details

Table 3-3: Operating modes overview

Operating Mode		Available sensor signals			Fusion Data	
		Accel	Mag	Gyro	Relative orientation	Absolute orientation
Non-fusion modes	CONFIGMODE	-	-	-	-	-
	ACCONLY	X	-	-	-	-
	MAGONLY	-	X	-	-	-
	GYROONLY	-	-	X	-	-
	ACCMAG	X	X	-	-	-
	ACCGYRO	X	-	X	-	-
	MAGGYRO	-	X	X	-	-
Fusion modes	AMG	X	X	X	-	-
	IMU	X	-	X	X	-
	COMPASS	X	X	-	-	X
	M4G	X	X	-	X	-
	NDOF_FMC_OFF	X	X	X	-	X
	NDOF	X	X	X	-	X

- UNIT_SEL : 0x3B

4.3.60 UNIT_SEL 0x3B

	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Access	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w
Reset	1	0	0	0	0	0	0	0
Content	ORI_Android_Windows	reserved		TEMP_Unit	reserved	EUL_Unit	GYR_Unit	ACC_Unit

Table 3-11: unit selection

Data	Units	[Reg Addr]: Register Value
Acceleration	m/s ²	[UNIT_SEL] : xxxxxxx0b
	mg	[UNIT_SEL] : xxxxxxx1b
Linear Acceleration, Gravity vector	m/s ²	[UNIT_SEL] : xxxxxxx0b
Magnetic Field Strength	Micro Tesla	NA
Angular Rate	Dps	[UNIT_SEL] : xxxxxx0xb
	Rps	[UNIT_SEL] : xxxxxx1xb
Euler Angles	Degrees	[UNIT_SEL] : xxxxx0xxb
	Radians	[UNIT_SEL] : xxxxx1xxb
Quaternion	Quaternion units	NA
Temperature	°C	[UNIT_SEL] : xxx0xxxxb
	°F	[UNIT_SEL] : xxx1xxxxb

Table 3-29: Euler angle data representation

Unit	Representation
Degrees	1 degree = 16 LSB
Radians	1 radian = 900 LSB

Table 3-33: Linear Acceleration data representation

Unit	Representation
m/s^2	1 m/s^2 = 100 LSB
mg	1 mg = 1 LSB

Table 3-37: Temperature data representation

Unit	Representation
$^{\circ}\text{C}$	1 $^{\circ}\text{C}$ = 1 LSB
F	2 F = 1 LSB

- Euler angels (0x1A ~ 0x1F)

1F	EUL_Pitch_MSB	0x00	Pitch Data <15:8>
1E	EUL_Pitch_LSB	0x00	Pitch Data <7:0>
1D	EUL_Roll_MSB	0x00	Roll Data <15:8>
1C	EUL_Roll_LSB	0x00	Roll Data <7:0>
1B	EUL_Heading_MSB	0x00	Heading Data <15:8>
1A	EUL_Heading_LSB	0x00	Heading Data <7:0>

- Quaternion (0x20 ~ 0x27)

27	QUA_Data_z_MSB	0x00	Quaternion z Data <15:8>
26	QUA_Data_z_LSB	0x00	Quaternion z Data <7:0>
25	QUA_Data_y_MSB	0x00	Quaternion y Data <15:8>
24	QUA_Data_y_LSB	0x00	Quaternion y Data <7:0>
23	QUA_Data_x_MSB	0x00	Quaternion x Data <15:8>
22	QUA_Data_x_LSB	0x00	Quaternion x Data <7:0>
21	QUA_Data_w_MSB	0x00	Quaternion w Data <15:8>
20	QUA_Data_w_LSB	0x00	Quaternion w Data <7:0>

- Accelerometer (0x8 ~ 0xD) : w/ Gravity

D	ACC_DATA_Z_MSB	0x00	Acceleration Data Z <15:8>
C	ACC_DATA_Z_LSB	0x00	Acceleration Data Z <7:0>
B	ACC_DATA_Y_MSB	0x00	Acceleration Data Y <15:8>
A	ACC_DATA_Y_LSB	0x00	Acceleration Data Y <7:0>
9	ACC_DATA_X_MSB	0x00	Acceleration Data X <15:8>
8	ACC_DATA_X_LSB	0x00	Acceleration Data X <7:0>

- Gyroscope (0x14 ~ 0x19)

19	GYR_DATA_Z_MSB	0x00	Gyroscope Data Z <15:8>
18	GYR_DATA_Z_LSB	0x00	Gyroscope Data Z <7:0>
17	GYR_DATA_Y_MSB	0x00	Gyroscope Data Y <15:8>
16	GYR_DATA_Y_LSB	0x00	Gyroscope Data Y <7:0>
15	GYR_DATA_X_MSB	0x00	Gyroscope Data X <15:8>
14	GYR_DATA_X_LSB	0x00	Gyroscope Data X <7:0>

- Accelerometer (0x28 ~ 0x2D) : w/o Gravity

2D	LIA_Data_Z_M BS	0x00	Linear Acceleration Data Z <15:8>
2C	LIA_Data_Z_LS B	0x00	Linear Acceleration Data Z <7:0>
2B	LIA_Data_Y_M BS	0x00	Linear Acceleration Data Y <15:8>
2A	LIA_Data_Y_LS B	0x00	Linear Acceleration Data Y <7:0>
29	LIA_Data_X_M BS	0x00	Linear Acceleration Data X <15:8>
28	LIA_Data_X_LS B	0x00	Linear Acceleration Data X <7:0>

- Temperature : 0x34

34	TEMP	0x00	Temperature
----	------	------	-------------

Code 수정

- UART1으로 로그를 내 보내기 위해 아래 코드 추가 :
core/src/main_example.c

```
/* printf → UART1 리디렉션 */  
int _write(int file, char *ptr, int len)  
{  
    HAL_UART_Transmit(&huart1, (uint8_t *)ptr, (uint16_t)len, 100);  
    return len;  
}
```

- BNO055의 slave address는 0x28이다.

```
/* _____  
 * I2C 주소  
 * COM3 핀 High → 0x29 (default)  
 * COM3 핀 Low → 0x28  
 * _____*/  
#define BNO055_I2C_ADDR_HIGH (0x29 << 1) /* COM3 = HIGH (default) */  
#define BNO055_I2C_ADDR_LOW (0x28 << 1) /* COM3 = LOW  
*/
```

- 초기화시에 Software Reset를 한다.

```
BNO055_Status_t BNO055_Init(BNO055_Handle_t *hbno,
                             I2C_HandleTypeDef *hi2c,
                             uint8_t i2c_addr,
                             bool use_ext_crystal)
{
    BNO055_Status_t ret;
    uint8_t chip_id;

    /* — CHIP_ID 확인 (0xA0 고정) — */
    ret = bno_read_reg(hbno, BNO055_CHIP_ID_REG, &chip_id);

    /* — 소프트웨어 리셋 — */
    ret = BNO055_Reset(hbno);

    /* — CONFIG 모드 진입 확인 — */
    ret = bno_set_mode_internal(hbno, BNO055_MODE_CONFIG);

    /* — 전원 모드: Normal — */
    ret = bno_write_reg(hbno, BNO055_PWR_MODE_REG, BNO055_POWER_NORMAL);

    /* — 단위 설정: m/s², dps, degree, °C, Windows 좌표계 —
       UNIT_SEL[7]=0: Windows [4]=0: °C
       [2]=0: degree [1]=0: dps [0]=0: m/s² */
    ret = bno_write_reg(hbno, BNO055_UNIT_SEL_REG, 0x00);

    /* — NDOF 모드 진입 (9축 Sensor Fusion) — */
    ret = bno_set_mode_internal(hbno, BNO055_MODE_NDOF);
    if (ret != BNO055_OK) return ret;

    return BNO055_OK;
}
```

Code 수정

- 1초 주기로 센서 데이터를 표시한다.

```
While (1) {
    /* 오일러각 (Fusion 출력) */
    if (BNO055_GetEuler(&bno, &euler) == BNO055_OK) {
        printf("Euler | Heading=%6.2f Roll=%6.2f Pitch=%6.2f [deg]\r\n",
            (double)euler.heading, (double)euler.roll, (double)euler.pitch);
    }

    /* 쿼터니언 (Fusion 출력) */
    if (BNO055_GetQuaternion(&bno, &quat) == BNO055_OK) {
        printf("Quat | W=%6.4f X=%6.4f Y=%6.4f Z=%6.4f\r\n", (double)quat.w, (double)quat.x, (double)quat.y,
            (double)quat.z);
    }

    /* 가속도 (중력 포함) */
    if (BNO055_GetAccel(&bno, &accel) == BNO055_OK) {
        printf("Accel | X=%6.2f Y=%6.2f Z=%6.2f [m/s^2]\r\n", (double)accel.x, (double)accel.y, (double)accel.z);
    }

    /* 자이로 */
    if (BNO055_GetGyro(&bno, &gyro) == BNO055_OK) {
        printf("Gyro | X=%7.2f Y=%7.2f Z=%7.2f [dps]\r\n", (double)gyro.x, (double)gyro.y, (double)gyro.z);
    }

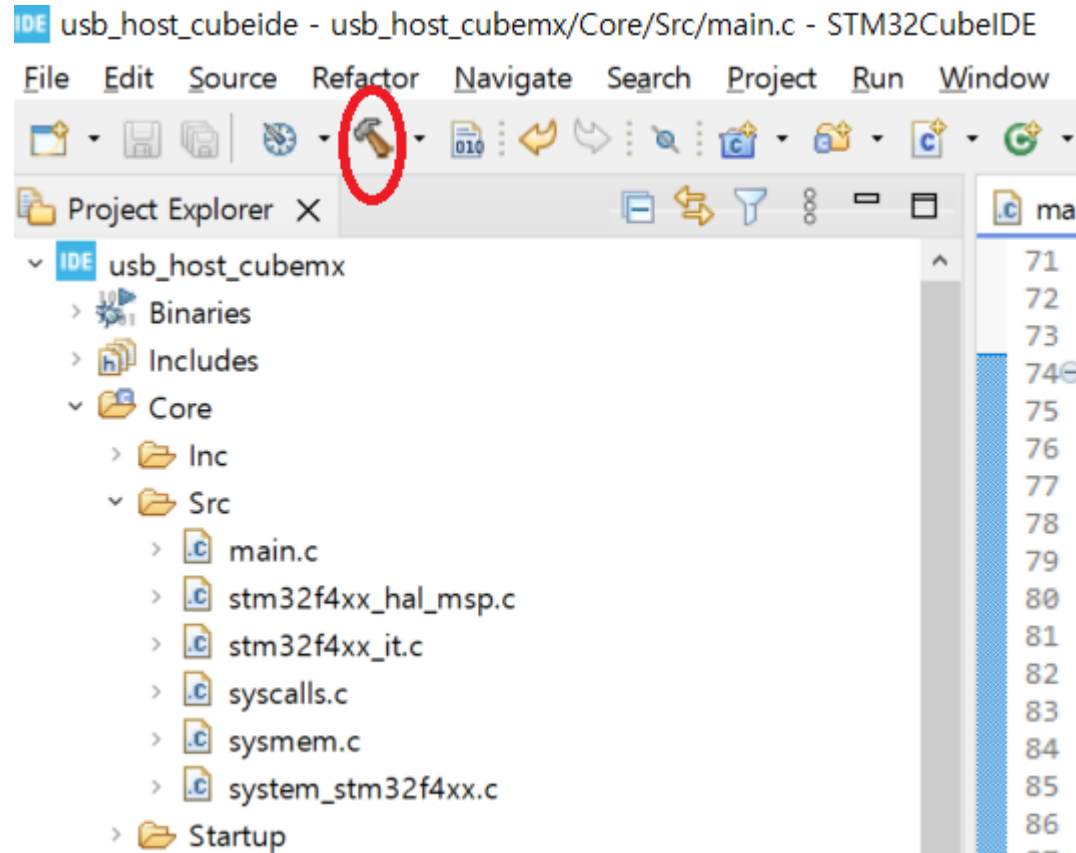
    /* 선형가속도 (중력 제거, Fusion 출력) */
    if (BNO055_GetLinearAccel(&bno, &lia) == BNO055_OK) {
        printf("LinAcc | X=%6.2f Y=%6.2f Z=%6.2f [m/s^2]\r\n", (double)lia.x, (double)lia.y, (double)lia.z);
    }

    /* 온도 */
    if (BNO055_GetTemperature(&bno, &temp) == BNO055_OK) {
        printf("Temp | %d [°C]\r\n", temp);
    }

    HAL_Delay(1000); /* 1 Hz 출력 */
}
```

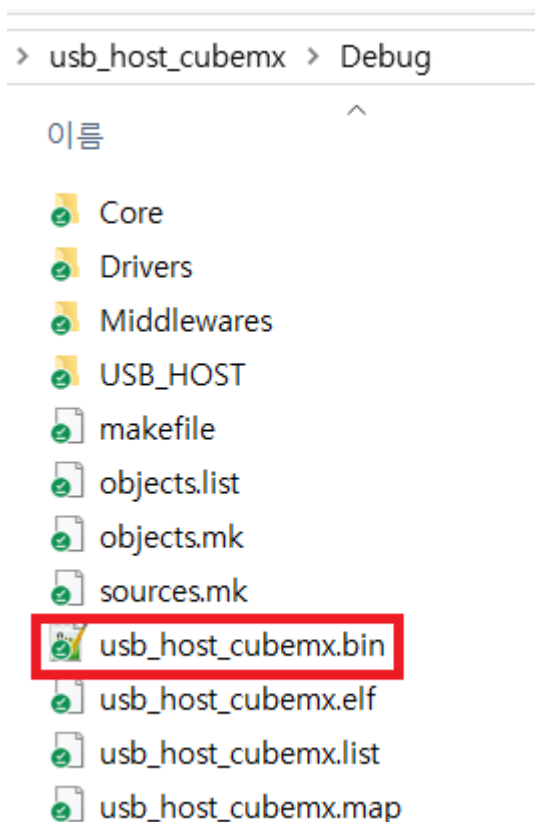
Build

- 아래 버튼 클릭하여 Build



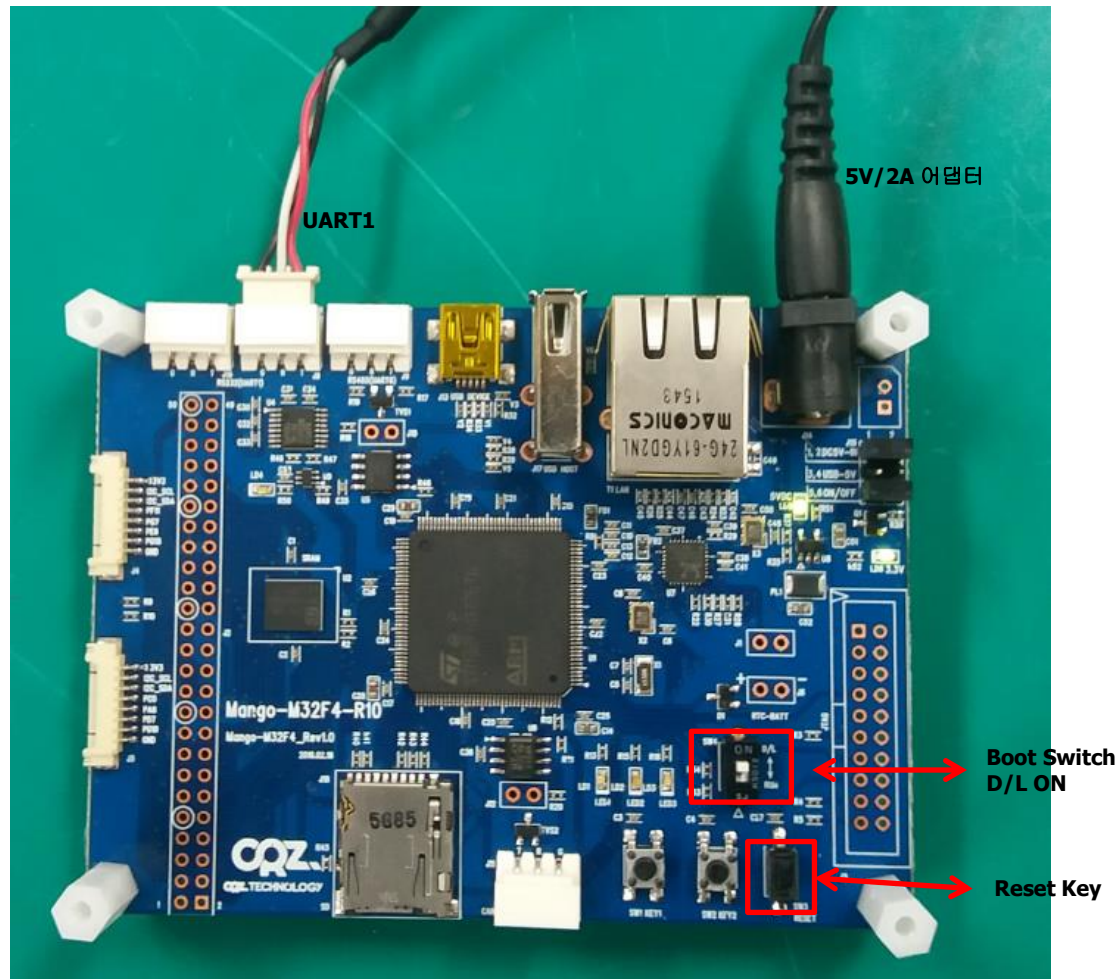
Build 결과물

- 빌드 결과물 이름은 [프로젝트 이름].bin이다.



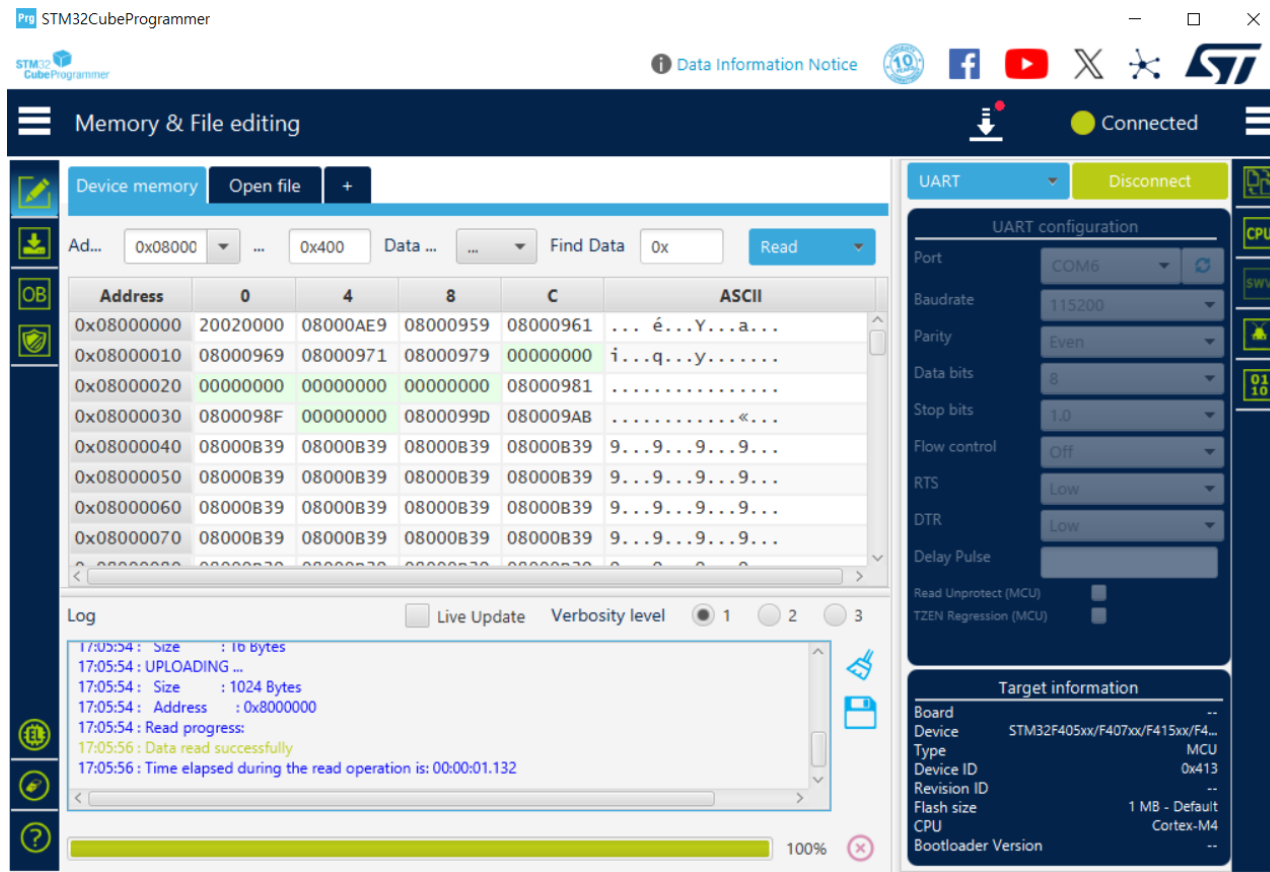
BIN 다운로드

- UART1과 연결되어 있는 터미널(Teraterm) 종료.
- Mango-M32F4를 다운로드 모드로 설정하고 SW3 리셋 키를 누른다.



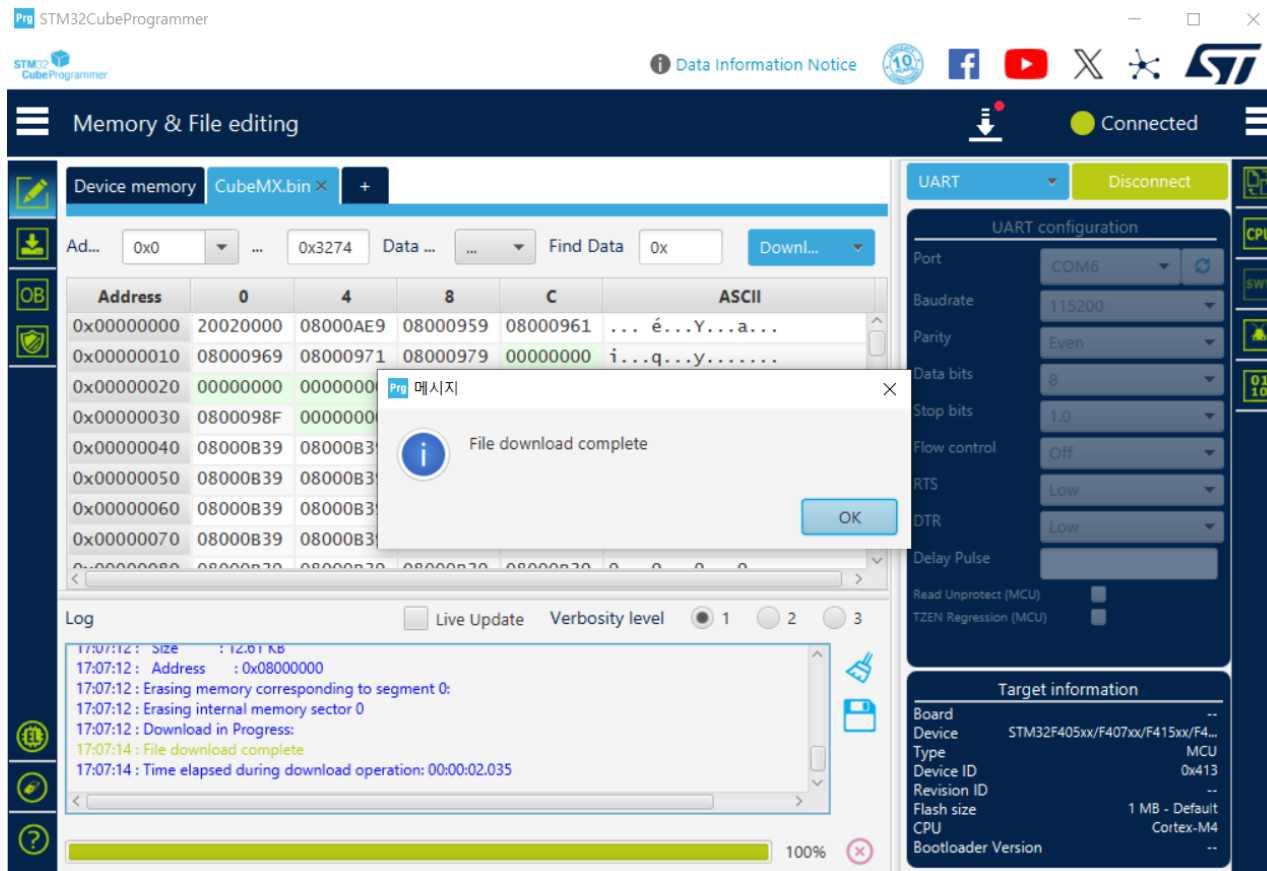
BIN 다운로드

- STM32CubeProgrammer를 실행한다.



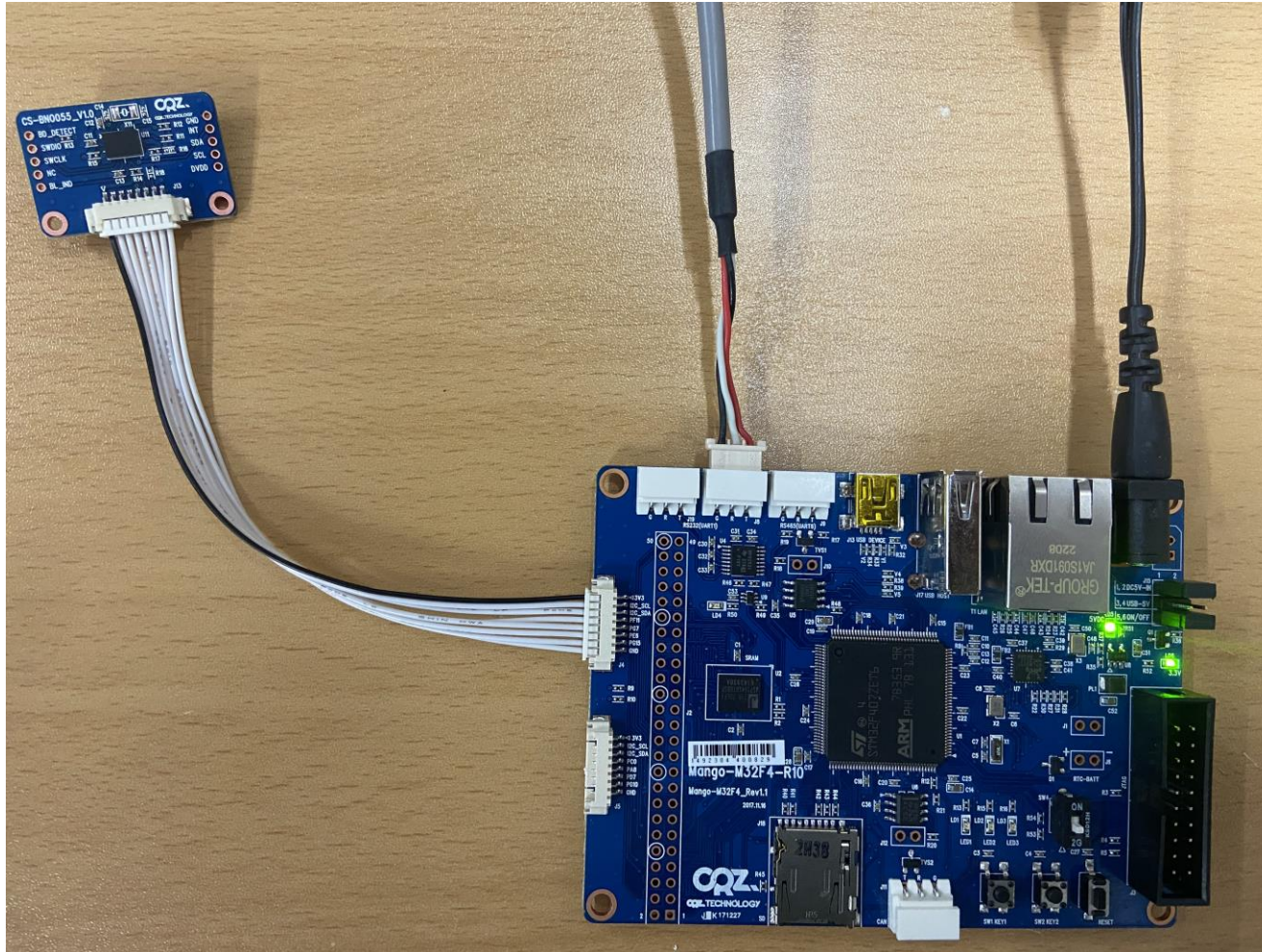
BIN 다운로드

- 빌드된 BIN 파일을 열어서 다운로드한다.



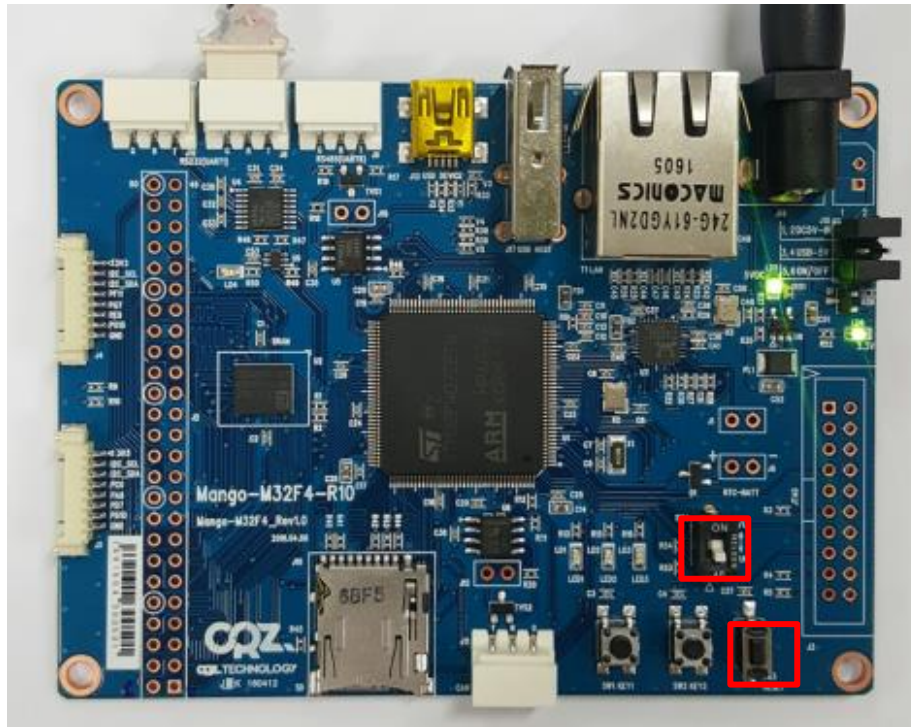
9축 센서 연결

- CS-BNO055 센서 보드를 Mango-M32F4 보드의 J4 혹은 J5에 8핀 커넥터를 이용하여 연결한다.



실행결과

- STM32CubeProgrammer의 COM Port를 연결해제하고 터미널 (Teraterm)을 실행한다.
- Mango-M32F4를 실행모드로 설정하고 SW3 리셋 키를 누른다.



- 9축 센서 데이터가 정상적으로 읽힘.

```
COM6 - Tera Term VT
File Edit Setup Control Window Help

===== BNO055 STM32F4 I2C Example =====
[OK] BNO055 initialized (NDOF mode)
SYS_STATUS=0x05  SYS_ERR=0x00  ST_RESULT=0x0F

--- Sensor Data ---
Euler | Heading= 0.00 Roll= 0.00 Pitch= 0.00 [deg]
Quat  | W=0.0000 X=0.0000 Y=0.0000 Z=0.0000
Accel | X= -1.62 Y= 9.29 Z= 2.62 [m/s²]
Gyro  | X= -0.25 Y= -0.12 Z= 0.31 [dps]
LinAcc| X= 0.00 Y= 0.00 Z= 0.00 [m/s²]
Temp  | 30 [°C]

Euler | Heading=359.94 Roll= -9.56 Pitch= 74.12 [deg]
Quat  | W=0.7968 X=0.5952 Y=0.1048 Z=0.0001
Accel | X= -1.56 Y= 8.87 Z= 2.48 [m/s²]
Gyro  | X= 0.19 Y= -0.19 Z= 0.00 [dps]
LinAcc| X= 0.08 Y= -0.43 Z= -0.15 [m/s²]
Temp  | 30 [°C]

Euler | Heading=359.94 Roll= -9.56 Pitch= 74.00 [deg]
Quat  | W=0.7973 X=0.5944 Y=0.1047 Z=0.0002
Accel | X= -1.62 Y= 9.26 Z= 2.61 [m/s²]
Gyro  | X= -0.06 Y= -0.31 Z= 0.19 [dps]
LinAcc| X= 0.02 Y= -0.03 Z= -0.03 [m/s²]
Temp  | 30 [°C]
```